Giuseppe **Attanasio**

Postdoctoral Researcher · Instituto de Telecomunicações

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"Computers aren't the thing. They are the thing that gets us to the thing." - Halt And Catch Fire

Bio

I'm a postdoc in the SARDINE group at Instituto de Telecomunicações, Lisbon, headed by scientific director André Martins. My research interests lie at the intersection of interpretability, fairness, and safety. Previously, I was a MilaNLP member, working with Debora Nozza and Dirk Hovy.

I am a computer engineer who cares about the ethical use of AI on people, communities, and society at large, about which I have given talks and published modest science. Some of the things I'm proud of from my past are co-founding a student team and RiTA, an online community connecting passionate people around Italian resources, and getting decently good at pizza-making. I find cross-institution and cross-discipline collaboration fun and formative. As I trust that deep understanding is rooted in hands-on work, I also enjoy coding for my research projects.

I update my website more frequently than this document, with recent professional news and projects.

Research Experience

Instituto de Telecomunicações — *Lisbon, Portugal*
Postdoc — *Feb. 2024 - ongoing*
PI: André Martins

Bocconi University — *Milan, Italy*
Postdoc at Department of Computing Sciences — *Oct. 2022 - Feb. 2024*
PI: Debora Nozza

Selected Publications

I firmly believe in open-access, reproducible research. Most of my papers are free to read, and code and data are open-source under a permissive license, whenever possible.

Attanasio, **Giuseppe**, Savoldi, Beatrice, Fucci, Dennis, and Hovy, Dirk. "Twists, Humps, and Pebbles: Multilingual Speech Recognition Models Exhibit Gender Performance Gaps". In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Ed. by Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 21318–21340. URL: `https://aclanthology.org/2024.emnlp-main.1188/`

Zaranis, Emmanouil, **Attanasio**, **Giuseppe**, Agrawal, Sweta, and Martins, Andre. "Watching the Watchers: Exposing Gender Disparities in Machine Translation Quality Estimation". In: *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar. Vienna, Austria: Association for Computational Linguistics, July 2025, pp. 25261–25284. ISBN: 979-8-89176-251-0. URL: `https://aclanthology.org/2025.acl-long.1228/`

Attanasio, **Giuseppe**, Plaza del Arco, Flor Miriam, Nozza, Debora, and Lauscher, Anne. "A Tale of Pronouns: Interpretability Informs Gender Bias Mitigation for Fairer Instruction-Tuned Machine Translation". In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. Ed. by Houda Bouamor, Juan Pino, and Kalika Bali. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 3996–4014. URL: `https://aclanthology.org/2023.emnlp-main.243`

Attanasio, **Giuseppe**, Nozza, Debora, Hovy, Dirk, and Baralis, Elena. "Entropy-based Attention Regularization Frees Unintended Bias Mitigation from Lists". In: *Findings of the Association for Computational*

Linguistics: ACL 2022. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 1105–1119. URL: https://aclanthology.org/2022.findings-acl.88

Attanasio, **Giuseppe**, Pastor, Eliana, Di Bonaventura, Chiara, and Nozza, Debora. "ferret: a Framework for Benchmarking Explainers on Transformers". In: *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*. Dubrovnik, Croatia: Association for Computational Linguistics, May 2023, pp. 256–266. URL: https://aclanthology.org/2023.eacl-demo.29

Delobelle, Pieter, **Attanasio**, **Giuseppe**, Nozza, Debora, Blodgett, Su Lin, and Talat, Zeerak. "Metrics for What, Metrics for Whom: Assessing Actionability of Bias Evaluation Metrics in NLP". in: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Ed. by Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 21669–21691. URL: https://aclanthology.org/2024.emnlp-main.1207/

Cercas Curry, Amanda, **Attanasio**, **Giuseppe**, Talat, Zeerak, and Hovy, Dirk. "Classist Tools: Social Class Correlates with Performance in NLP". in: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Lun-Wei Ku, Andre Martins, and Vivek Srikumar. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 12643–12655. URL: https://aclanthology.org/2024.acl-long.682/

Bianchi, Federico, Suzgun, Mirac, **Attanasio**, **Giuseppe**, Rottger, Paul, Jurafsky, Dan, Hashimoto, Tatsunori, and Zou, James. "Safety-Tuned LLaMAs: Lessons From Improving the Safety of Large Language Models that Follow Instructions". In: *The Twelfth International Conference on Learning Representations*. 2024. URL: https://openreview.net/forum?id=gT5hALch9z

Röttger, Paul, Kirk, Hannah, Vidgen, Bertie, **Attanasio**, **Giuseppe**, Bianchi, Federico, and Hovy, Dirk. "XSTest: A Test Suite for Identifying Exaggerated Safety Behaviours in Large Language Models". In: *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Ed. by Kevin Duh, Helena Gomez, and Steven Bethard. Mexico City, Mexico: Association for Computational Linguistics, June 2024, pp. 5377–5400. URL: https://aclanthology.org/2024.naacl-long.301/

Please, head to my Google Scholar page or visit https://gattanasio.cc/uploads/pubs.pdf for the full list.

Grants & Awards

Grants

2025	**EuroHPC Extreme Scale Access**, Project: SMURF4EU: A Suite of Multimodal Reasoning Foundation Models for Europe	*Co-PI, EuroHPC*
2025	**FCT Advanced Computing**, Project: MELT: Multilingual Efficient Learning for Speech-Text Models. Funds: €39,000	*FCT*
2025	**EAMT Sponsorship**, Project: A User-Centered Benchmark for Usability Assessment of Speech Translation Systems. Funds: €5,200	*EAMT*
2023	**EAMT Sponsorship**, Project: GeFMT: Gender-Fair Language in German Machine Translation. Funds: €5,200	*EAMT*
2023	**ISCRA-C**, Project: ItaLLM: Advancing Beyond-English Language Models: A Computational Framework for Italian Language Modeling. Funds: 10,200 GPU hours	*CINECA*
2023	**Google Cloud Credits**, Project: Gender Bias in Large Language Models. Funds: €4,800	*Google*

Awards

2024	**Social Impact Paper Award**, Paper: Twists, Humps, and Pebbles: Multilingual Speech Recognition Models Exhibit Gender Performance Gaps	*EMNLP*
2023	**Best Student Paper Award**, Paper: Contrastive Language–Image Pre-training for the Italian	*CLiC-it*

Impact and Titles

2025	**ASN**, Italy's Abilitazione Scientifica Nazionale, 01/B1 Informatica, Seconda Fascia
2024	**10M+ downloads**, FashionCLIP at `https://huggingface.co/patrickjohncyh/fashion-clip`
2024	**30K+ downloads**, ferret at `https://github.com/g8a9/ferret`
2024	**20K+ downloads**, simple-generation at `https://github.com/MilaNLProc/simple-generation`

Service Work

Workshop Organization

- **(Expected) 2026**: I will co-organize the next edition of the Gender Bias in NLP workshop.
- **2024**: I co-organized the third edition of the Third Workshop on Safety for Conversational AI during LREC COLING 2024.
- **2022**: I co-organized the first edition of EvalRS: a Rounded Evaluation of Recommender Systems, a data challenge and workshop held during CIKM 2022.

Editorial

- **Area Chair**: ACL Rolling Review, LREC/COLING
- **Senior Area Chair**: LREC/COLING.

Peer Reviewing. I reviewed at least one work submitted to the following venues: ACL Rolling Review (2021-onward); ACM KDD SIGKDD Conference On Knowledge Discovery And Data Mining (2020, 2021); ACM SIGMOD/PODS International Conference on Management of Data (2021); IEEE ICDE: IEEE International Conference on Data Engineering (2020); IEEE ICDM: IEEE International Conference on Data Mining (2021); ACM SAC: ACM/SIGAPP Symposium On Applied Computing (2021); EDBT: International Conference on Extending Database Technology (2020, 2021); DaWaK: International Conference on Big Data Analytics and Knowledge Discovery (2019, 2021).

I reviewed at least one work submitted to the following journals: Expert Systems With Applications, Elsevier; Future Generation Computer Systems, Elsevier; Machine Learning With Applications, Elsevier.

Industry and Academic Collaborations

I have engaged in several fruitful research collaborations with industry partners, leading to impactful outcomes in fairness, inclusivity, multimodality, and recommender systems.

- **Fairness in End-to-End Speech Representation Models** — Studied subgroup biases in neural speech models (e.g., Wav2Vec 2.0, HuBERT) to detect underperformance on underrepresented groups. *Entities: Amazon Alexa AI.*
- **Gender Inclusive Communication in Italian** — Developed methods for detecting and rephrasing non-inclusive language in Italian public and academic texts. *Entities: University of Turin, University of Bologna, University of Bergamo.*
- **Contrastive Language-Image Pre-training for Italian** — Fine-tuned CLIP for Italian, producing the first Italian multimodal CLIP model, finalist at HuggingFace Flax/JAX Community Week. *Entities: Bocconi University, University of Milano-Bicocca, University of Groningen.*
- **EvalRS: Rounded Evaluation of Recommender Systems** — Proposed EvalRS, an open-source framework and challenge addressing fairness and behavioral aspects in RecSys evaluation (CIKM 2022). *Entities: Coveo, Coveo Labs, Stanford University, Microsoft, NVIDIA.*
- **Vision and Language for Fashion Concept Learning** — Demonstrated that CLIP-based multimodal models efficiently capture general fashion concepts (e.g., sleeve length, heel type). *Entities: Coveo, Coveo Labs, Stanford University, Farfetch.*

Teaching

Teaching Assistant. I have been a teaching assistant for several courses at Politecnico di Torino. In total, I have held 171 hours of complementary teaching. The list of courses where I TAd is:

(2021) **Data science lab: processes and methods**, 60h, MD in Data Science and Engineering; (2020) **Data science lab: processes and methods**, 39h, MD in Data Science and Engineering; (2020) **Business Intelligence for Big Data**, 21h, MD in Industrial engineering and management; (2019) *** Data science lab: processes and methods**, 39h, MD in Data Science and Engineering; (2019) **Business Intelligence for Big Data**, 21h, MD in Industrial engineering and management; (2018) **Basi di dati**, 21h, BD in Computer Engineering; (2018) **The Fourth Industrial Revolution: Promises and Pitfalls in Blending New and Traditional Approaches in Manufacturing and Service Sectors**, 30h, Alta Scuola Politecnica School.

* I played a key role in launching Data Science Lab: Process and Methods for the Data Science and Engineering Master's at Politecnico. The course introduces Python and core Data Science and Machine Learning libraries, with 10 labs providing 60+ pages of exercises and 250+ pages of solutions.

Tutoring. I co-tutored the second edition of the **Mediterranean Machine Learning (M2L) Summer School**, with ownership of the Natural Language Processing part. We prepared a shared notebook to guide students through 1) implementing a Transformer from scratch, 2) training the model for language modeling and machine translation, and 3) testing on gender bias. I have also tutored the hands-one laboratories during the 2024 **LxMLS Machine Learning School** in Lisbon. I co-founded **MAchine Learning At poliTO** (MALTO), a university-funded student team at Politecnico di Torino. The team's primary goal is to participate in various data science competitions. I have supervised the work of more than **10 master's students**.

Guest Lecturer.

- 2023, Bocconi University: **Contrastive Learning**
- 2023, Bocconi University: **Reinforcement Learning from Human Feedback**
- 2023, Tilbug University: **Social Biases in LMs: Why They Are There, How Do We Measure and Mitigate Them**

Industry & Associations

I have held the following courses:

- (2024) **Generative Artificial Intelligence and Basic Prompting: How to Train AI Tools for Work in Publishing Houses**
  AIE: Associazione Italiana Editori (online)
- (2021) **Data Theory: Data Visualization with Python**
  Reply: Digital Services, Technology and Consulting, Turin (online)
- (2020) **Python technologies for Data Analytics**
  Applied Mechatronic Engineering & Technologies, Turin

Invited Talks

I have disseminated my research through several invited talks. Some of the host institutions and entities were: University of Padova, the Italian Association of Editors, Cambridge Forum of Science and Humanities, Karger Publishers, Papers We Love (Milan's Chapter), and SmartData@PoliTO.

Education

Politecnico di Torino *Turin, Italy*
Ph.D. at Department of Control and Computer Engineering *Nov. 2018 - Oct. 2022*
Advisor: Elena Baralis

Politecnico di Torino *Turin, Italy*
MD in Computer Engineering, Data Science track *Oct. 2016 - Oct. 2018*
Grade: 110/110 *cum Laude*

Politecnico di Torino *Turin, Italy*
BD in Computer Engineering *Oct. 2013 - Oct. 2016*
Grade: 110/110

Skills

As I frequently pursue and practice new interests, this list evolves quickly. I am currently out of practice with *italicized* entries meaning that I am not actively using them but can quickly regain proficiency with some dedicated effort.

Programming Python, *C, C++, Java, C#, JavaScript, PHP, Matlab, SQL*
Deep Learning PyTorch, *JAX*

Graphic Design Inkscape, *GIMP*

Languages Italian, English, *Portuguese, Spanish, French*